

Lecture 7 & 8

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E1 248
Sliding mode control and its application

Introduction

1. VIBRATIONAL CONTROL OF AEROPLANE D.C. GENERATOR (KULEBAKIN V., 1932)

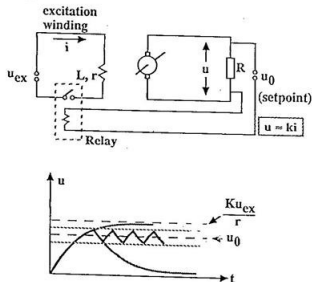


Figure: Primitive Examples - Electrical

2. ON AUTOMATIC STABILITY OF A SHIP ON A GIVEN COURSE (NIKOLSKI G., 1934)

φ - SHIP COURSE

δ - RUDDER POSITION

U - ON-OFF CONTROL WITH DEAD ZONE

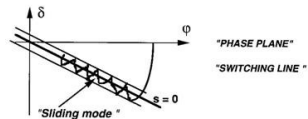
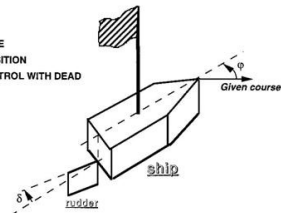


Figure: Primitive Examples-Mechanical

Sliding mode control: Revisited

Variable Structure Control: Prof. Emelyanov

- The first steps of sliding mode control theory originated in early 1950 and were initiated by S. V. Emelyanov.
- Started as Variable Structure Control (VSC):-Varying system structure for stabilization.

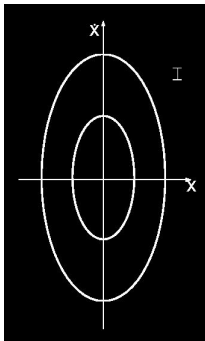


Figure: Mode I

$$\ddot{x} = -k_2 x$$

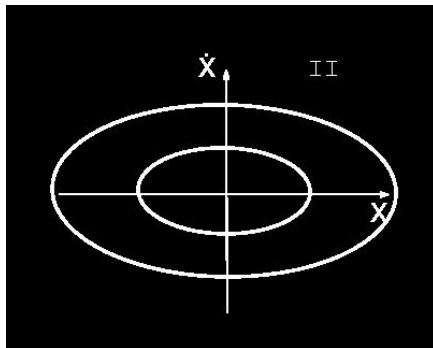
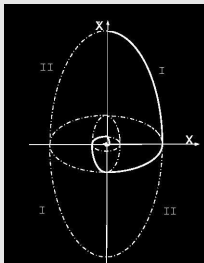


Figure: Mode II $\ddot{x} = -k_1 \dot{x}$, $0 < k_1 < 1 < k_2$

Sliding mode control: Revisited

Piecing together



Properties of VSC

- Both constituent systems were oscillatory and were not asymptotically stable.
- Combined system is asymptotically stable.
- Property not present in any of the constituent systems is obtained by VSC.

Sliding mode control: Revisited

Unstable Constituent Systems

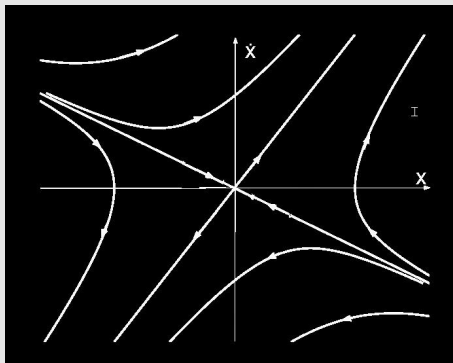


Figure: $\ddot{x} - \xi\dot{x} - \alpha x = 0$

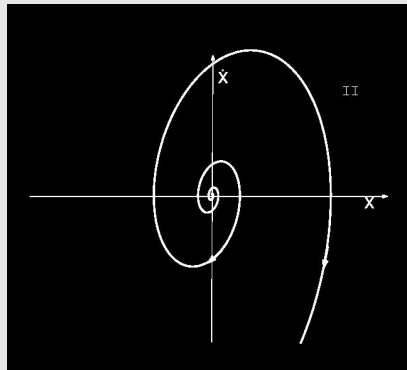


Figure: $\ddot{x} - \xi\dot{x} + \alpha x = 0$

Sliding mode control: Revisited

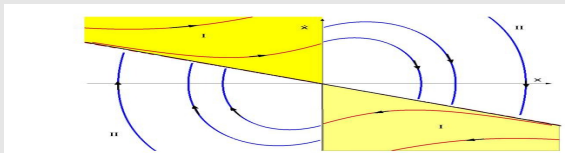
Analysis

- Both systems are unstable
- Only stable mode is one mode of the system

$$\ddot{x} - \xi \dot{x} - \alpha x = 0, \lambda = \frac{\xi}{2} - \sqrt{\frac{\xi^2}{4} + \alpha}$$

- If the following VSC is employed

$$\text{Mode} = \begin{cases} I & \text{if } xs \leq 0 \\ II & \text{if } xs > 0. \end{cases}, s = cx + \dot{x}, c = -\lambda$$



- Again, property not present in constituent systems is found in the combined system, and a stable structure can be obtained by varying between two unstable structures.

Sliding mode control design

Consider the linear system

$$\dot{x} = Ax + B(u + d(t, x))$$

$$x \in \mathbb{R}^n, u \in \mathbb{R}.$$

- The sliding surface is designed as $\sigma(x) = Cx = 0$.
- The sliding mode control is designed to reach the sliding surface in finite time.

Assumptions

- The pair (A, B) , is controllable.
- The scalar $CB \neq 0$.
- The disturbance/uncertainties are matched and bounded, i.e., $|d(t, x)| \leq D_M$.

Sliding mode control design

Controller Design

Since the function $\sigma(x)$ is to be made zero, it helps to look at its derivative.

$$\begin{aligned}\dot{\sigma} &= C\dot{x} \\ &= CAx + CB(u + d(t, x))\end{aligned}\tag{1}$$

Now, let us design the control as,

$$u = (CB)^{-1} (-CAx + \phi(\sigma)).$$

Substituting this control in (1) gives,

$$\dot{\sigma} = \phi(\sigma) + CBd(t, x)\tag{2}$$

- The function $\phi(\sigma)$ must be selected such that, the equation (2) provides finite time convergence of σ to zero, in presence of $d(t, x)$.
- Once the sliding surface is reached, i.e., $\sigma(x) \equiv 0$, the system trajectory is governed by zero dynamics, also called reduced-order dynamics.

Sliding mode control design

Examples of $\phi(\sigma)$

- ① $\phi(\sigma) = -K \operatorname{sgn}(\sigma)$
- ② $\phi(\sigma) = -\lambda\sigma - K \operatorname{sgn}(\sigma)$
- ③ $\phi(\sigma) = -K|\sigma|^\alpha \operatorname{sgn}(\sigma)$

However, the third choice can not reject the disturbances if $d(t, x) \neq 0$ at $x = 0$.

Relative Degree Problem

$$\dot{\sigma} = CAx + CB(u + d(t, x))$$

- If $CB = 0$, then control does not appear in the expression for $\dot{\sigma}$.
- Usually, C is designed surface matrix so $CB = 0$ can be avoided.

However, σ may be some output of the system, and C is determined by system dynamics. The objective is to make the output zero robustly, then with $CB = 0$, the sliding mode control design is not possible.

Chattering Problem

The control is discontinuous, resulting in high-frequency switching by the actuator.

Invariance in Sliding-modes

B. Drazenovic "The invariance conditions in variable structure systems" Automatica, v.5, No.3, Pergamon Press, 1969.



Figure: Prof. B. Drazenovic

Mathematical Treatment of Matched Uncertainty

Mathematical meaning of equivalent control is that, when the trajectories are on the sliding surface then σ and $\dot{\sigma}$, both must be zero. But this assumption is valid only when the control input appears linearly for a given state. Now consider the following system

$$\dot{x} = f(x) + g(x)u + d \quad (3)$$

Invariance in Sliding modes

Consequently, one can write

$$\dot{\sigma} = \frac{\partial \sigma}{\partial x} \dot{x} = \frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial x} g(x)u + \frac{\partial \sigma}{\partial x} d \quad (4)$$

For calculating equivalent value of control substitute $\dot{\sigma} = 0$ in Eqn.(4), the only condition is that $\frac{\partial \sigma}{\partial x} g(x) \neq 0$, we can write

$$u_{\text{equivalent}} = - \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \left(\frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial x} d \right) \quad (5)$$

The closed loop dynamics during sliding is obtained by substituting (5) into (3), so

$$\begin{aligned} \dot{x} &= f(x) - g(x) \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \left(\frac{\partial \sigma}{\partial x} f(x) + \frac{\partial \sigma}{\partial x} d \right) + d \\ &= \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \frac{\partial \sigma}{\partial x} \right) f(x) + \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \frac{\partial \sigma}{\partial x} \right) d \end{aligned} \quad (6)$$

Invariance in Sliding-modes

- Now suppose that disturbance d is entering through control channel.
- Therefore one can write $d = g(x)\delta(t)$, where $\delta(t)$ is unknown signal, but with a known bound.
- This kind of disturbance is known as matched disturbance.
- After substituting the value of d , into (6), one can write

$$\begin{aligned}
 \dot{x} &= \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \frac{\partial \sigma}{\partial x} \right) f(x) + \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \frac{\partial \sigma}{\partial x} \right) g(x)\delta(t) \\
 &= \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \frac{\partial \sigma}{\partial x} \right) f(x) + g(x)\delta(t) - g(x)\delta(t) \\
 &= \left(I - g(x) \left(\frac{\partial \sigma}{\partial x} g(x) \right)^{-1} \frac{\partial \sigma}{\partial x} \right) f(x)
 \end{aligned} \tag{7}$$

- Hence the closed loop dynamics, which is given in (7), is completely independent of any matched disturbances when the system is in the sliding mode.
- This property is known as the **invariance property** with respect to disturbance.

Sliding mode control design using regular form transformation

Consider the linear time-invariant system of the following form

$$\dot{x}(t) = (A + \Delta A(t))x(t) + Bu + Dd(t) \quad (8)$$

where $x \in \mathbb{R}^{n \times 1}$, $u \in \mathbb{R}^{m \times 1}$, $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ are the state vector, control input, system matrix and input matrix respectively.

$\Delta A(t)$ and $d(t)$ represent parameter variations and disturbance vector, where D is the disturbance input matrix.

Assumptions

- The system is assumed to be controllable, i.e., the controllability matrix $[B \ AB \ A^2B \ \dots \ A^{n-1}B]$ has full rank.
- $\Delta A \in \text{range}(B)$ and $D \in \text{range}(B)$, or there exist constant or time-varying matrices Ψ_1 and Ψ_2 , such that

$$\Delta A = B\Psi_1, \quad D = B\Psi_2 \quad (9)$$

Using Eqn.(8) and assumption (9), one can write

$$\dot{x}(t) = Ax(t) + B(u + \underbrace{\Psi_1 x(t) + \Psi_2 d(t)}_{\Psi_0}) = Ax(t) + B(u + \Psi_0), \quad \Psi_0 = \Psi_1 x(t) + \Psi_2 d(t)$$

Sliding mode control design using regular form transformation

Regular Form Transformation:

Input matrix B in Eqn. (8) may be partitioned (after reordering the state vector components) as

$$B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

where $B_1 \in \mathbb{R}^{(n-m) \times m}$, $B_2 \in \mathbb{R}^{m \times m}$ with $\det B_2 \neq 0$.

The non-singular coordinate transformation T , to convert system (8) into regular form is given by

$$T = \begin{bmatrix} I_{n-m} & -B_1 B_2^{-1} \\ 0 & B_2^{-1} \end{bmatrix}$$

change of coordinate is

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = T x$$

Sliding mode control design using regular form transformation

Transformed system (8), after applying above change of coordinate

$$\begin{aligned}\dot{z}_1(t) &= A_{11}z_1(t) + A_{12}z_2(t) \\ \dot{z}_2(t) &= A_{21}z_1(t) + A_{22}z_2(t) + u + \Psi_0\end{aligned}\tag{10}$$

where $z_1 \in \mathbb{R}^{n-m}$, $z_2 \in \mathbb{R}^m$ and

$$TAT^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad TB = \begin{bmatrix} 0 \\ I \end{bmatrix}.$$

Design of Sliding Surface

- Now, our next aim is to design a specified switching manifold such that after reaching the manifold, the system is free from its dynamics.
- The designed manifold can be linear or nonlinear, depending on our specified goal. The simplest manifold is a linear one.
- So, here, we focus on linear manifolds only, unless otherwise specified.d.

Sliding mode control design using regular form transformation

The linear switching variable (sliding variable) is

$$S = C_1 z_1 + z_2 \quad (11)$$

where $S \in \mathbb{R}^{m \times 1}$ and $C_1 \in \mathbb{R}^{m \times (n-m)} > 0$.

Taking the first time derivative of the sliding surface (11) and substituting the value from the Eqn.(10), one can get

$$\begin{aligned} \dot{S} &= C_1 \dot{z}_1 + \dot{z}_2 \\ &= (C_1 A_{11} + A_{21})z_1(t) + (C_1 A_{12} + A_{22})z_2(t) + u + \Psi_0 \end{aligned} \quad (12)$$

Expressing the system (10) into z_1 and S coordinate, using the differential Eqns.(11) and (12)

$$\begin{aligned} \dot{z}_1(t) &= A_{11}z_1(t) + A_{12}(S - C_1 z_1(t)) \\ \dot{S} &= (C_1 A_{11} + A_{21})z_1(t) + (C_1 A_{12} + A_{22})(S - C_1 z_1(t)) + u + \Psi_0 \\ &= (C_1 A_{11} + A_{21} - C_1 A_{12} C_1 - A_{22} C_1)z_1(t) + (C_1 A_{12} + A_{22})S + u + \Psi_0 \end{aligned} \quad (13)$$

Sliding mode control design using regular form transformation

Existence Condition of Sliding Mode:- The main objective here is to design a control u such that the sliding motion occurs in finite time. For this, the control is selected according to the following theorem

Theorem

The control input u which is defined as

$$u = \nu - (C_1 A_{11} + A_{21} - C_1 A_{12} C_1 - A_{22} C_1) z_1(t) - (C_1 A_{12} + A_{22}) S \quad (14)$$

where,

$$\nu = -K_1 S - K_2 \text{sign}(S)$$

with $K_1 > 0$, $K_2 > \Psi_M = \max \|\Psi_0\|$ leads to the establishment S equal to zero in finite time.

Proof

Substituting u from Eqn.(14) to (13), one can write

$$\begin{aligned} \dot{z}_1(t) &= A_{11} z_1(t) + A_{12}(S - C_1 z_1(t)) \\ \dot{S} &= -K_1 S - K_2 \text{sign}(S) + \Psi_0 \end{aligned} \quad (15)$$

Sliding mode control design using regular form transformation

Choosing the Lyapunov function as $V = \frac{1}{2} S^T S$, and calculating the time derivative of the Lyapunov function along (15), one can get

$$\begin{aligned}\dot{V} &= S^T \dot{S} = S^T (-K_1 S - K_2 \text{sign}(S)) + \Psi_0 \\ &= -K_1 V - K_2 \|S\| + S^T \Psi_0\end{aligned}$$

Using Cauchy-Schwartz inequality $S^T \Psi_0 \leq \|S\| \|\Psi_0\|$ and $(2V)^{\frac{1}{2}} = (S^T S)^{\frac{1}{2}} = \|S\|$, so

$$\begin{aligned}\dot{V} &\leq -K_1 V - (K_2 - \|\Psi_0\|) \|S\| \\ &\leq -K_1 V - (K_2 - \Psi_M) \|S\| \\ &= -K_1 V - 2^{\frac{1}{2}} (K_2 - \Psi_M) V^{\frac{1}{2}} \\ &= -K_1 V - \eta V^{\frac{1}{2}} < 0\end{aligned}$$

Therefore, one can ensure the asymptotic stability of the trajectory to the sliding manifold where $\eta = 2^{\frac{1}{2}} (K_2 - \Psi_M)$.

Sliding mode control design using regular form transformation

Now, we have to prove that convergence takes place in finite time.

Time calculation:-

$$\dot{V} = \frac{dV}{dt} \leq -k_1 V - \eta V^{\frac{1}{2}}$$

Use comparison lemma

$$\dot{u} = -k_1 u - \eta u^{\frac{1}{2}}$$

Then

$$T \leq \frac{2}{k_1} \left(\ln \left(k_1 V(0)^{\frac{1}{2}} + \eta \right) - \ln(\eta) \right)$$

Therefore, the above claim is justified since the time T is always finite.

Equivalent Control

It is clear from (14), during the sliding mode $S = 0$ and the average value of $\nu = 0$, therefore the remaining control is $u_{\text{equivalent}} = -(C_1 A_{11} + A_{21} - C_1 A_{12} C_1 - A_{22} C_1) z_1(t)$, which is theoretically interpreted as **equivalent control**, which is the control required to maintain a sliding motion on the manifold S .

Sliding mode control design using regular form transformation

Analysis of Reduced Order Dynamics

- Therefore, after time T , dynamics of S have been collapsed, and the reduced order dynamics of the system (15) can be obtained by substituting $S = 0$.
- The reduced order dynamics is given as

$$\dot{z}_1(t) = (A_{11} - A_{12}C_1)z_1(t)$$

- The stability and performance of the above reduced order system depend on (A_{11}, A_{12}) .
- Therefore, the design of C_1 depends on the controllability of the pair (A_{11}, A_{12}) .

Lemma

The matrix pair (A_{11}, A_{12}) is controllable, if and only if the pair (A, B) is controllable.

Sliding mode control design using regular form transformation

Proof

$$\begin{aligned}\text{rank}[sI - A \ B] &= \text{rank} \begin{bmatrix} sI - A_{11} & -A_{12} & 0 \\ -A_{21} & sI - A_{22} & I \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} sI - A_{11} & A_{12} \end{bmatrix} + m \text{ for all } s \in \mathbb{C}\end{aligned}$$

This implies

$$\text{rank}[sI - A \ B] = n \Leftrightarrow \text{rank}[sI - A_{11} \ A_{12}] = n - m$$

and from the Popov-Belevitch-Hautus (PBH) rank test, it follows that (A, B) is controllable if and only if the pair (A_{11}, A_{12}) is controllable.

Now, one can design C_1 using any robust linear state feedback method, such as quadratic minimization, direct or robust eigenvalue assignment, using LMI, etc.

Thank You!